# ROTABLATION-ASSISTED LEFT MAIN PCI

## A single centre experience

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## BACKGROUND

# Calcified left main disease makes PCI hard, yet for some patients PCI is the only option.

# Why Indian data matter

- Extensive, multivessel, heavily calcified CAD
- CAD a decade earlier than in the West
- High burden of diabetes and dyslipidemia
- DAPT compliance hit by cost and awareness
- Little rotablation LM-PCI evidence from India

# Prospective study design

**45**

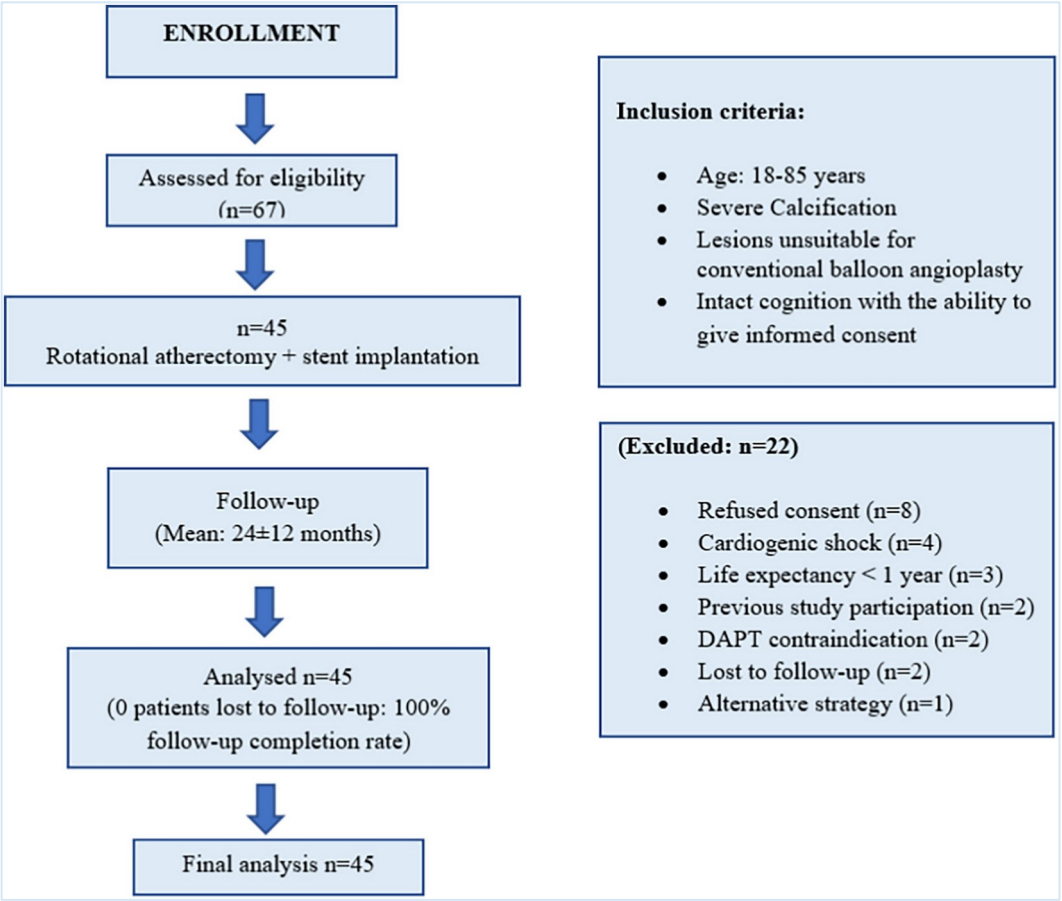
consecutive patients, rota LM-PCI

**2020–23**

enrolled Jan 2020 – Jan 2023

**2 years**

follow-up, 100% complete




# Patient selection (Fig. 1)

Heart team: SYNTAX >32 advised CABG

PCI by patient choice after counselling

LMCA ≥50% + severe Ca or IVUS Ca score ≥2

# Rotablator RA technique

| 0.7 | 1.25–2.0 | 140–180k |
| --- | --- | --- |
| target burr : artery ratio | mm burrs; multiple if needed | rpm; burr runs <30 s |

# Outcome definitions

## In-hospital

- Angiographic: <20% residual, TIMI 3
- Procedural: no major complication, death, MI or emergency revasc.

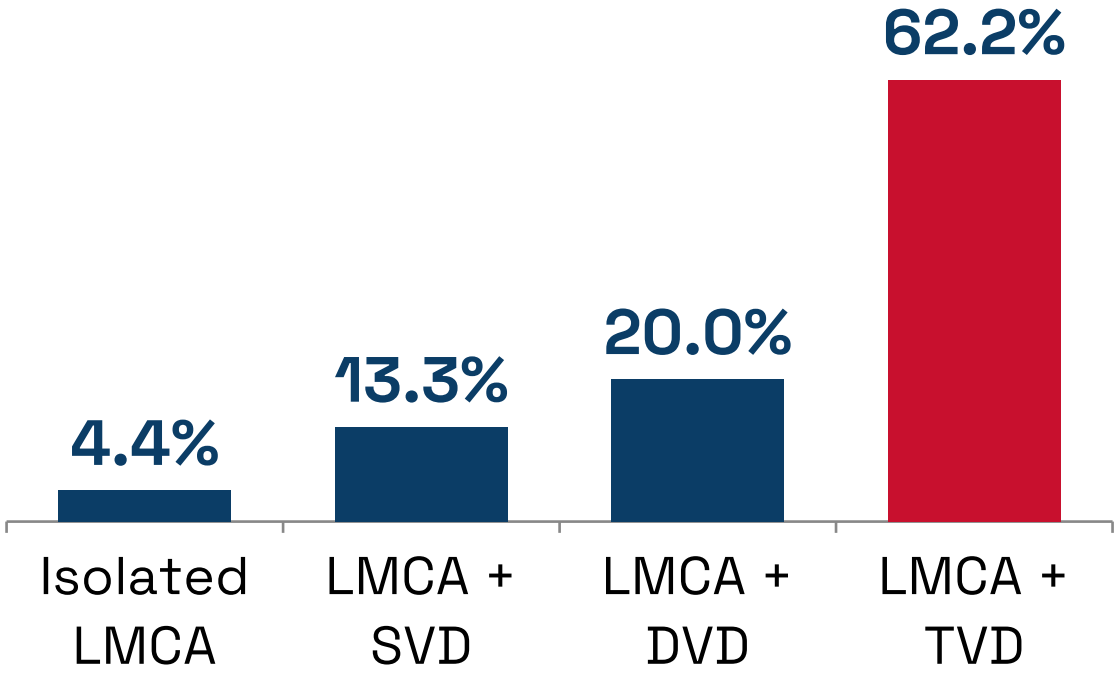
## Two-year

- MACE: cardiac death, nonfatal MI, TLR
- Stent thrombosis (ARC), stroke, bleeding (BARC)

# Elderly, high-risk cohort

| Clinical profile | Value | Risk / history | n (%) |
|---|---|---|---|
| Age (years) | 75.2 ± 5.9 | Hypertension | 38 (84.4%) |
| Male | 31 (68.9%) | Diabetes | 25 (55.6%) |
| LVEF (%) | 43.3 ± 8.1 | Previous PCI | 15 (33.3%) |
| SYNTAX score | 40.1 ± 8.2 | Previous CABG | 11 (24.4%) |
| EuroSCORE II | 5.4 ± 1.9 | NSTEMI | 24 (53.3%) |

# Mostly LM + 3-vessel CAD




**86.7%**
severe LMCA calcification

**62.2%**
distal LMCA lesion

# Procedural details

| Technique | n (%) | Parameter | Value |
|---|---|---|---|
| Femoral access | 38 (84.4%) | Max burr, mm | 1.50 ± 0.13 |
| 7F guide | 45 (100%) | Burrs/patient | 1.11 ± 0.31 |
| Two-stent | 34 (75.6%) | Stent len., mm | 22.98 ± 6.54 |
| DK crush | 32 (71.1%) | Stent dia., mm | 3.49 ± 0.40 |
| IVUS | 34 (75.6%) | IABP, n (%) | 7 (15.6%) |

# In-hospital outcomes

**100%**

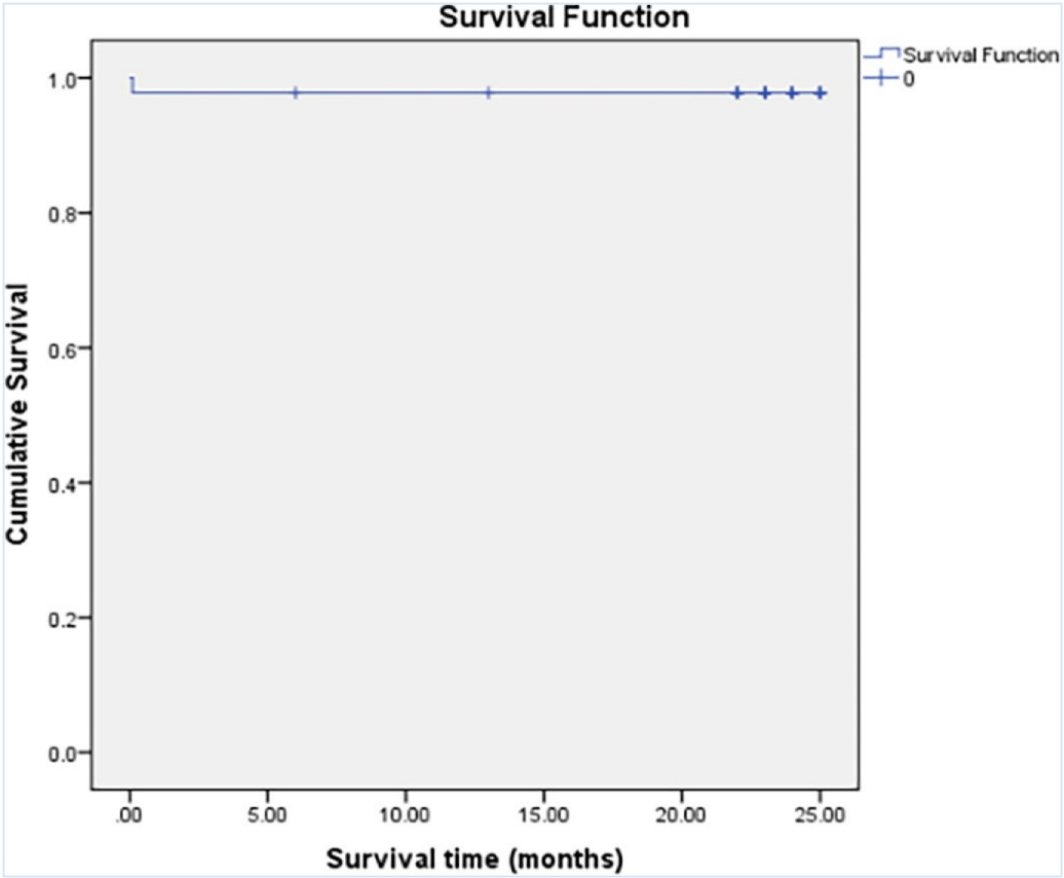
angiographic success (45/45)

**97.8%**

procedural success (44/45)

**2.2%**

in-hospital death (1/45)




# Two-year outcomes

Overall survival **97.78%**
(Fig. 2, Kaplan-Meier)

MACE-free **93.3%**

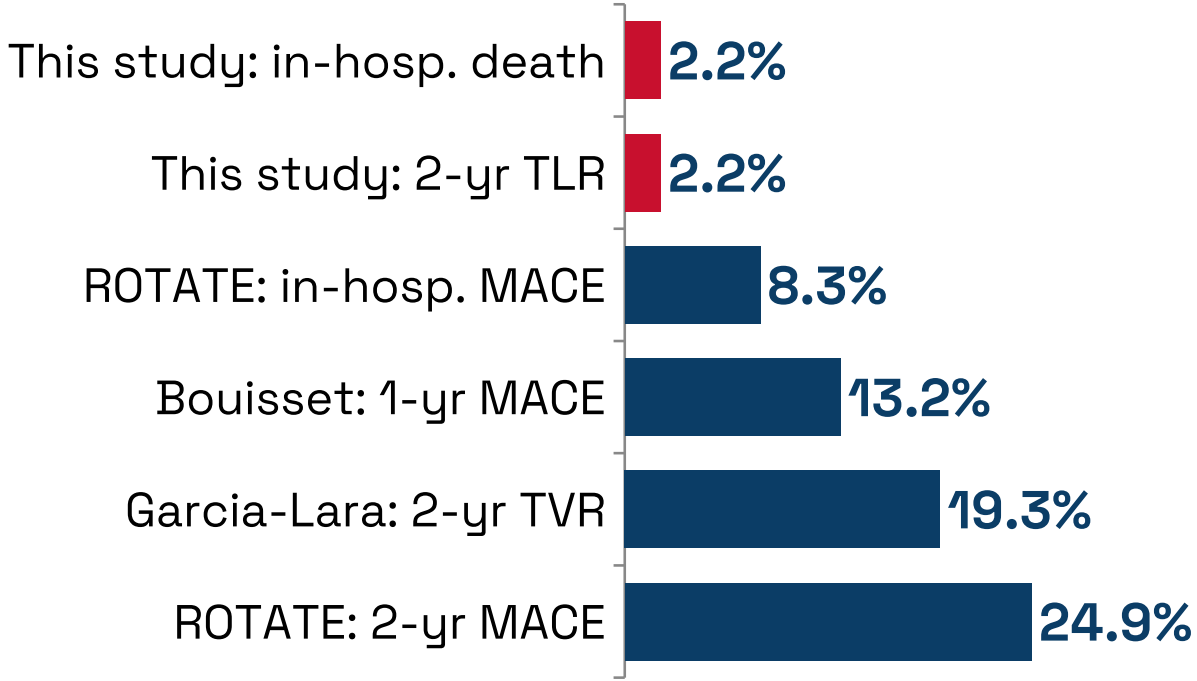
1 death, 1 TLR, 1 TVMI

No bleeding or stroke

# Correlation analysis

| Variable pair | Pearson r | 95% CI | p |
|---|---|---|---|
| Age vs SYNTAX score | 0.23 | 0.02–0.42 | 0.028 |
| EuroSCORE II vs age | 0.52 | 0.34–0.67 | <0.001 |
| LMCA stenosis vs fluoro | 0.31 | 0.11–0.49 | 0.003 |
| Contrast vs fluoro time | 0.38 | 0.19–0.55 | 0.001 |
| Burr size vs stent diam. | 0.45 | 0.26–0.61 | <0.001 |
| SYNTAX vs stent length | 0.28 | 0.08–0.46 | 0.006 |

# Endpoints differ by study




**Not like-for-like**

**Garcia-Lara 2-yr cardiac-mortality-free survival**

**71 ± 7%**

# Limitations

## Design

- Single centre, small sample (n = 45)
- No control group (non-rota PCI or CABG)

## Scope

- No systematic angiographic follow-up
- Rotablator RA only: not orbital or IVL

# Take-home messages

- Rota-assisted LM-PCI is feasible, safe and effective in severely calcified lesions
- Procedural success 97.8%, with a low rate of periprocedural complications
- Durable two-year outcomes: MACE-free survival 93.3%

# THANK YOU

Questions and discussion

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